

Xiaoxuan HEI

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EDUCATION

ENSTA, Institut Polytechnique de Paris

Ph.D. in Computer Science; Supervisor: Prof. Adriana Tapus

Palaiseau, France

Oct. 2022 – June 2026

Université Paris-Saclay

Master in Computer Science (Data & Knowledge)

Orsay, France

Sept. 2019 – Sept. 2020

Polytech Paris-Saclay

Engineer in Computer Science

Orsay, France

Sept. 2017 – Sept. 2020

Huazhong University of Science and Technology

Bachelor in Optoelectronic Information

Wuhan, China

Sept. 2014 – June 2018

EXPERIENCE

Big Data Engineer

New Oriental Education & Technology Group

Jan. 2021 – Aug. 2022

Beijing, China

- Designed and implemented a buried point (event tracking) system for user behavior data collection.
- Built a Change Data Capture (CDC) pipeline to synchronize and process real-time data streams.
- Developed and maintained Data Warehouse services to support analytics and business intelligence.

Research Intern

ENSTA, Institut Polytechnique de Paris

May 2020 – Sept. 2020

Palaiseau, France

- Speech-Driven Robot Face Generation Project: Trained a deep generative model on paired speech audio and facial expression data to enable robots to generate facial expressions from input speech.

Interpreter Intern

8th Arrondissement Police Station

Aug. 2018 – Sept. 2018

Paris, France

- Provided Chinese–French interpretation, assisting Chinese speakers in filing police reports.

PUBLICATIONS

- Lu, Z., Shangguan, Z., **Hei, X.**, Benoit, C., Stephenson, M., and Tapus, A. (2026), “Supervising Autonomous Ships: How Display Modality and Encounter Geometry Shape Trust and Situation Awareness”, In 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, PA, USA, September 2026
- **Hei, X.**, and Tapus, A. (2026) “A POMDP-Based Framework for Humor-Aware Human-Robot Dialogue”, In 35th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Kitakyushu, Japan, August 2026
- **Hei, X.**, Gioumatzidou, S., García Cárdenas, J.J., and Tapus, A. (2026) “Investigating Cross-Agent Trust Transfer: Effects of Informational and Operational Errors in Human-Robot Collaboration”, In Companion of the 2026 ACM/IEEE International Conference on Human-Robot Interaction (HRI), Edinburgh, Scotland, UK, March 2026
- **Hei, X.**, Zhang, H., and Tapus, A. (2025), “Investigating the Impact of Humor on Learning in Robot-Assisted Education”, In 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, October 2025
- Shangguan, Z., **Hei, X.**, Li, F., Yu, C., Song, S., Zhao, J., Cangelosi, A., and Tapus, A. (2025), “Learning from Human Conversations: A Seq2Seq based Multi-modal Robot Facial Expression Reaction Framework in HRI”, In 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, October 2025 (Best Paper Award Finalist)
- **Hei, X.**, Zhang, H. and Tapus, A. (2025) “Estimating User Engagement in Human Robot Interaction Using a Dynamic Bayesian Network”, In 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, USA, May 2025
- Zhang, H., Saood, A., García Cárdenas, J.J., **Hei, X.**, and Tapus, A. (2025) ““Oh! It’s Fun Chatting with You!” A Humor-aware Social Robot Chat Framework”, In 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, USA, May 2025

- García Cárdenas, J.J., **Hei, X.**, and Tapus, A. (2024) “Exploring Cognitive Load Dynamics in Human-Machine Interaction for Teleoperation: A User-Centric Perspective on Remote Operation System Design”, In 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Abu Dhabi, UAE, October 2024
- **Hei, X.**, Zhang, H., and Tapus, A. (2024) “Exploring Help-Seeking Behavior, Performance, and Cognitive Load in Individual Tutoring: A Comparative Study between Human Tutors and Social Robots”, In 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Pasadena, California, USA, August 2024
- Zhang, H., **Hei, X.**, Zhong, J., Tapus, A. (2024) “Robot Laughter: Does an appropriate laugh facilitate the robot’s humor performance?”, In 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Pasadena, California, USA, August 2024
- Zhang, H., **Hei, X.**, García Cárdenas, J.J., Miao, X., Tapus, A. (2024) “Toward a Multi-dimensional Humor Dataset for Social Robots”, In 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Pasadena, California, USA, August 2024
- **Hei, X.**, Zhang, H., Yu, C., and Tapus, A. (2024) “A bilingual social robot with sign language and natural language”, In Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI), Boulder, Colorado, USA, March 2024
- Zhang, H., Yu, C., **Hei, X.**, and Tapus, A. (2024) “Semantic Gesture in Robot Humor: A New Dimension to Enhance the Humor Expression”, In Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI), Boulder, Colorado, USA, March 2024
- **Hei, X.**, Denis, V., Oréface, P.H., Afyouni, A., Laborde, P., Legois, D., Ocnareescu, I., Anastassova, M., Tapus, A. (2023) “Evaluating Students’ Experiences in Hybrid Learning Environments: A Comparative Analysis of Kubi and Double Telepresence Robots”, In International Conference of Social Robotics (ICSR), Doha, Qatar, December 2023
- **Hei, X.**, Zhang, H., and Tapus, A. (2023) “Robots in education: Influence of Regulatory Focus Theory”, In 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Busan, South Korea, August 2023
- Yu, C., Zhang, H., Shangguan, Z., **Hei, X.**, Cangelosi, A., and Tapus, A. (2022) “Speech-driven Robot Face Action Generation with Deep Generative Model for Social Robots”, In International Conference of Social Robotics (ICSR), Florence, Italy, December 2022 (Best Paper Award Finalist)

TEACHING EXPERIENCE

Teaching Assistant

ENSTA, Institut Polytechnique de Paris

- ROB 311: Learning for robotics (2022, 2023, 2025)
- IN 104: C language project (2023)

AWARDS

- Travel Award for RO-MAN 2026 August 2026
- IROS-SDC Travel Award for IROS 2025 October 2025
- IROS 2025 Social Media Ambassador of IEEE Robotics and Automation Society October 2025
- RAS Travel Grant for ICRA 2025 May 2025
- Best Poster Award at Welcome Day of the Doctoral School IP Paris 2023 - 2024 December 2023
- Honor of Outstanding Graduates of HUST June 2018
- Honor of Excellent Student of HUST May 2017

PROFESSIONAL SERVICE

Reviewer for Conferences

- ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2024, 2026
- IEEE International Conference on Robotics and Automation (ICRA) 2024, 2025, 2026
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024, 2025, 2026
- IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) 2024, 2025, 2026
- IEEE International Conference on Development and Learning (ICDL) 2025
- International Conference of Social Robotics (ICSR) 2025
- Towards Autonomous Robotic Systems Conference (TAROS) 2025

Reviewer for Journals

- Advanced Robotics Journal 2024
- ACM Transactions on Human-Robot Interaction (THRI) 2026
- Frontiers in Robotics and AI 2026

Organization

- RO-MAN 2026 Workshop “Human State-Aware Robotics (H-STAR): From Multimodal Data to Human-Adaptive Behavior in HRI”
- ICSR 2026 Workshop “2nd Workshop on Cognitive Load and Engagement in Human-Robot Collaboration: From Robots in Hazardous Environments to Educational Applications”
- IROS 2025 Workshop “Enhancing Human Engagement in Social Assistive Robotics: Exploring Interaction, Cognitive Load, and Adaptive Support”
- ICSR 2025 Workshop “Cognitive Load and Engagement in Human-Robot Collaboration: From Robots in Hazardous Environments to Educational Applications”

PROFESSIONAL SKILLS

- Languages: Chinese - Native; English - Fluent; French - Proficient; Korean - Proficient
- Programming Languages: Python, JAVA, C/C++
- Operating Systems: Linux (Ubuntu), MacOS, ROS (Robot Operating System)